C# & .NET Exam Paper 1: Core Fundamentals & Control Flow

Comprehensive Test covering .NET Architecture, IDE, Variables, Operators, and Loops

Duration: *90 Minutes* | ***Total Marks:*** *70 Marks* | ***Format:*** *Written & Practical Coding*

Section A: .NET Architecture & Visual Studio IDE (15 Marks)

Test your fundamental knowledge of the runtime, execution model, and development environment.

Q1. What is the primary role of the Common Language Runtime (CLR) in the .NET Framework? [3 Marks]

A) To edit C# source code files

B) To manage memory, execute C# code, and handle thread management

C) To store project files in Visual Studio

D) To convert HTML pages into C# classes

Q2. Explain the difference between Source Code, Intermediate Language (IL/MSIL), and Native Machine Code during C# compilation. [6 Marks]

Q3. In Visual Studio Community, what is the purpose of the Solution Explorer window and the Build solution command? [6 Marks]

Section B: Variables, Data Types & Operators (20 Marks)

Focuses on type safety, implicit/explicit casting, and mathematical/logical operators.

Q4. Which of the following data types is a C# value type stored directly on the Stack? [3 Marks]

A) string

B) int

C) List

D) string[]

Q5. Analyze the following C# code snippet. What will be printed to the Console? [7 Marks]

```
int a = 15;
int b = 4;
double result1 = a / b;
double result2 = (double)a / b;
Console.WriteLine($"{result1} | {result2}");
```

Q6. Write a short C# program that prompts the user to enter a temperature in Celsius via Console.ReadLine(), converts the input string to a double, calculates Fahrenheit (F = C * 9/5 + 32), and displays the formatted result. [10 Marks]

Section C: Control Flow & Iteration (35 Marks)

Evaluates conditional decision making and loop structures.

Q7. Convert the following if-else block into an equivalent switch statement: [10 Marks]

```
if (grade == "A") {
Console.WriteLine("Excellent");
} else if (grade == "B") {
Console.WriteLine("Good");
} else {
Console.WriteLine("Needs Improvement");
}
```

Q8. Write a C# program using a while loop that continuously prompts the user to enter numbers until they type -1. The program should sum all positive numbers entered and output the final sum. [25 Marks]

EXAM ANSWER KEY & MARKING SCHEME

Comprehensive Solutions & Grading Criteria

Solutions: Section A: .NET Architecture & Visual Studio IDE (15 Marks)

Q1. Solution & Criteria:

Correct Answer: B. The CLR is the execution engine of .NET that handles garbage collection, JIT compilation, memory management, and security enforcement.

Q2. Solution & Criteria:

Source code (.cs) is written by developers. The C# compiler (csc) converts source code into Intermediate Language (IL/CIL) stored in assemblies (.dll/.exe). At runtime, the JIT (Just-In-Time) compiler converts IL into CPU-specific native machine code.

Q3. Solution & Criteria:

Solution Explorer displays the hierarchical structure of projects, source code files, dependencies, and references. The Build solution command compiles all modified source files into IL binaries to detect syntax and reference errors.

Solutions: Section B: Variables, Data Types & Operators (20 Marks)

Q4. Solution & Criteria:

Correct Answer: B. Primitive numeric types like int, double, bool, and char are value types stored on the stack.

Q5. Solution & Criteria:

Output: 3 | 3.75
Explanation: a / b performs integer division resulting in 3. (double)a casts 15 to 15.0, forcing floating-point division resulting in 3.75.

Q6. Solution & Criteria:

```
Console.Write("Enter Celsius: ");
string input = Console.ReadLine();
double celsius = Convert.ToDouble(input);
double fahrenheit = (celsius * 9.0 / 5.0) + 32;
Console.WriteLine($"Fahrenheit: {fahrenheit:F2}");
```

Solutions: Section C: Control Flow & Iteration (35 Marks)

Q7. Solution & Criteria:

```
switch (grade) {
case "A":
Console.WriteLine("Excellent");
break;
case "B":
Console.WriteLine("Good");
break;
default:
Console.WriteLine("Needs Improvement");
break;
}
```

Q8. Solution & Criteria:

```
int totalSum = 0;
while (true) {
Console.Write("Enter a number (-1 to stop): ");
int num = int.Parse(Console.ReadLine());
if (num == -1) break;
if (num > 0) totalSum += num;
}
Console.WriteLine($"Total Sum of Positive Numbers: {totalSum}");
```